

***Generation Interconnection Request
System Impact Study Report
For
PJM Generation Interconnection Request
Queue Position AE1-198

Crete 345 kV***

December 2021

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study. The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

General

The Interconnection Customer (IC) has proposed a solar generating facility located in McHenry County, Illinois. The installed facilities will have a total capability of 357 MW with 193 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is 9/01/2021, per Attachment N. This study does not imply a TO commitment to this in-service date.

Point of Interconnection

Queue Position AE1-198, a 357 MW solar farm facility, proposes to interconnect with the ComEd transmission system at 138kV bus at Kewanee TSS 74. The existing Kewanee substation cannot be expanded to interconnect AE1-198 due to physical limitations. A new 138kV substation Kentville Road TSS 962 is proposed to be built adjacent to the Kewanee substation to interconnect a prior queue position AD1-031. It is proposed to interconnect AE1-198 to TSS 962 Kentville Road.

Cost Summary

The AE1-198 project will be responsible for the following costs.

Description	Cost Estimate
Total Physical Interconnection Costs	\$13,000,000
Allocation towards System Network Upgrade Costs (PJM Identified - Summer Peak)*	\$0
Allocation towards System Network Upgrade Costs (PJM Identified - Light Load)*	\$10,300,000
Allocation towards System Network Upgrade Costs (TO Identified)*	\$0
Total Costs	\$23,300,000

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If

your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required

Transmission Owner Scope of Work

The total physical interconnection costs is given in the table below:

Description	Cost Estimate
Attachment Facilities	\$1,000,000
Direct Connection Network Upgrades	\$12,000,000
Non-Direct Connection Network Upgrades	\$0
Total Physical Interconnection Costs	\$13,000,000

Attachment Facilities:

The AE1-198 generator lead will interconnect to 345kV bus at TSS 945 Crete (see details in Direct Connection section below). The required Attachment Facilities are one 345kV line MODs, one dead-end structure and one revenue-metering as shown in the one-line diagram.

Scope of Work	Cost Estimate
Installation of one 138kV line MOD, one dead-end structure and one set of revenue metering (see notes below on cost estimate)	\$1,000,000

Direct Connection Cost Estimate:

In order to accommodate interconnection of AE1-198, Crete TSS 945 would need to be expanded to create a bus position.

The scope of work includes installation of four 345kV circuit breakers, to create a line position for AE1-198 generator lead, as shown in the one-line diagram below.

The Interconnection Customer (“IC”) is responsible for constructing all the facilities on the Interconnection Customer side of the Point of Interconnection (POI).

The preliminary cost estimate for Direct Connection Network Upgrade is given in the following tables.

Scope of Work	Cost Estimate
Installation of four 345kV circuit breakers at 345kV TSS 945 Crete.	\$12,000,000

Non-Direct Connection Cost Estimate

None

Notes on Cost Estimate:

- 1) These estimates are Order-of-Magnitude estimates of the costs that ComEd would bill to the customer for this interconnection. These estimates are based on a one-line electrical diagram of the project and the information provided by the Interconnection Customer.
- 2) There were no site visits performed for these estimates. There may be costs related to specific site related issues that are not identified in these estimates. The site reviews will be performed during the Facilities Study or during detailed engineering.
- 3) These estimates are not a guarantee of the maximum amount payable by the Interconnection Customer and the actual costs of ComEd's work may differ significantly from these estimates. Per the PJM Tariff, Interconnection Customer will be responsible for paying all actual costs of ComEd's work.
- 4) The Interconnection Customer is responsible for all engineering, procurement, testing and construction of all equipment on the Interconnection Customer's side of the Point of Interconnection (POI).

Schedule:

Normally it takes about 24-months to engineer, design, procure material and construct 345kV facilities after ISA/ICSA are signed.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Irradiance (Watts/meter²) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Ambient air temperature (Fahrenheit) - (Accepted, not required)
- Wind speed (meters/second) - (Accepted, not required)
- Wind direction (decimal degrees from true north) - (Accepted, not required)

Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

Network Impacts

The Queue Project AE1-198 was evaluated as a 358 MW (Capacity 193 MW) injection at Crete 345 kV substation in the ComEd area. Project AE1-198 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-198 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

Overload Number	Type	Contingency		Affected Area	Bus		Circuit	Power Flow	Loading %		Rating Type	MV A	MW Contribution	Flowgate Appendix
		Name	Facility Description		From	To			Initial	Final				
1	DCTL	COMED_P7_345-L94507_B-S_+_345-L97008_R-S	E FRANKFO; B-GOODINGS ;3B 345 kV line	CE - CE	270728	270766	1	AC	97.86	109.74		2084	255.42	1

Note:

Overload 1: ComEd ALDR rating is 2397 MVA (Not a violation)

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

Overload Number	Type	Contingency Name	Affected Area	Facility Description	Bus		Circuit	Power Flow	Loading %		Rating Type	MV A	MW Contribution	Flowgate Appendix
					From	To			Initial	Final				
2	LFF B	AEP_P4_#2978_05DUMONT 765_B	MISO NIPS - CE	17GREEN_ACRE-GREENACRE; T 345 kV line	2551 04	2707 71	1	AC	100.16	100.54	ER	109 1	59.94	2
3	LFF B	COMED_P4_023-65-BT2-3__	MISO NIPS - CE	17STJOHN-ST JOHN ; T 345 kV line	2551 12	2708 86	1	AC	122.97	132.32	ER	109 1	104.3	3
4	LFF B	AEP_P4_#2978_05DUMONT 765_B	MISO NIPS - CE	17STJOHN-ST JOHN ; T 345 kV line	2551 12	2708 86	1	DC	122.61	132.2	ER	109 1	103.93	
5	LFF B	COMED_P4_112-65-BT4-5__	MISO NIPS - CE	17STJOHN-ST JOHN ; T 345 kV line	2551 12	2708 86	1	DC	121.94	131.57	ER	109 1	104.3	
6	LFF B	COMED_P4_112-65-BT3-4__	MISO NIPS - CE	17STJOHN-ST JOHN ; T 345 kV line	2551 12	2708 86	1	DC	121.94	131.57	ER	109 1	104.3	
7	LFF B	AEP_P4_#2978_05DUMONT 765_B	MISO NIPS - AEP	17STILLWELL-05DUMONT 345 kV line	2551 13	2432 19	1	AC	126.68	127.81	ER	140 9	56.56	4
8	BUS	COMED_P2-2_116_GG-345R_2_FSA	CE - CE	GOODINGS ;4B-GOODINGS ;3B 345 kV line	2707 70	2707 66	1	DC	104.82	106.25	LD R	180 2	25.4	5
9	LFF B	COMED_P4_116-45-L11614_	CE - CE	GOODINGS ;4B-GOODINGS ;3B 345 kV line	2707 70	2707 66	1	DC	102.34	103.79	LD R	180 2	25.76	
10	LFF B	COMED_P4_116-45-L9801_FSA	CE - CE	GOODINGS ;4B-GOODINGS ;3B 345 kV line	2707 70	2707 66	1	DC	100.82	102.24	LD R	180 2	25.4	
11	BUS	COMED_P2-2_116_GG-345R_2_NO_FSA	CE - CE	GOODINGS ;4B-GOODINGS ;3B 345 kV line	2707 70	2707 66	1	DC	100.82	102.24	LD R	180 2	25.4	
12	LFF B	AEP_P4_#2978_05DUMONT 765_B	CE - AEP	GREENACRE; T-05OLIVE 345 kV line	2707 71	2432 29	1	AC	112.54	112.96	ER	971	59.94	6
13	LFF B	COMED_P4_023-65-BT2-3__	CE - MISO NIPS	ST JOHN ; T-17GREEN_ACRE 345 kV line	2708 86	2551 04	1	AC	122.96	132.31	ER	109 1	104.3	7
14	LFF B	AEP_P4_#2978_05DUMONT 765_B	CE - MISO NIPS	ST JOHN ; T-17GREEN_ACRE 345 kV line	2708 86	2551 04	1	DC	122.6	132.19	ER	109 1	103.93	
15	LFF B	COMED_P4_112-65-BT4-5__	CE - MISO NIPS	ST JOHN ; T-17GREEN_ACRE 345 kV line	2708 86	2551 04	1	DC	121.94	131.57	ER	109 1	104.3	
16	LFF B	COMED_P4_112-65-BT3-4__	CE - MISO NIPS	ST JOHN ; T-17GREEN_ACRE 345 kV line	2708 86	2551 04	1	DC	121.94	131.57	ER	109 1	104.3	
17	LFF B	COMED_P4_112-65-BT4-5__	CE - MISO NIPS	CRETE EC ;BP-17STJOHN 345 kV line	2747 50	2551 12	1	AC	149.16	162.01	ER	139 9	178.46	8

18	LFF B	COMED_P4_112-65-BT3-4	CE - MISO NIPS	CRETE EC ;BP-17STJOHN 345 kV line	2747 50	2551 12	1	AC	149. 16	162. 01	ER	139 9	178.46	
19	LFF B	AEP_P4_#2978_05DUMONT 765_B	CE - MISO NIPS	CRETE EC ;BP-17STJOHN 345 kV line	2747 50	2551 12	1	AC	147. 08	159. 9	ER	139 9	178.12	

Note:

Overload 2: ComEd SSTE rating is 1134 MVA (Not a violation) MISO-end ratings are 1195/1195 MVA SN/SE (Not a violation)

Overload 3: MISO STE: 1591 MVA (No violation). ComEd SSTE: 1134 MVA (violation valid).

Overload 4: MISO STE: 1591 MVA (No violation). ComEd SSTE: 1134 MVA (violation valid)

Overload 5: MISO STE: 1591 MVA (No violation). ComEd SSTE: 1134 MVA (violation valid)

Overload 6: MISO STE: 1591 MVA (No violation). ComEd SSTE: 1134 MVA (violation valid)

Overload 7: AEP SE rating is 1409 MVA (Violation Valid) MISO ratings are 1409/1779 MVA (Violation Valid)

Overload 8: ComEd SSTE: 2083 MVA (No violation).

Overload 9: ComEd SSTE: 2083 MVA (No violation).

Overload 10: ComEd SSTE: 2083 MVA (No violation).

Overload 11: ComEd SSTE: 2083 MVA (No violation).

Overload 12: ComEd SSTE: 1134 MVA (No violation). AEP: 971 MVA SE (Violation Valid).

Overload 13: MISO: 1313/1591 MVA. ComEd SSTE: 1134 MVA (Violation for ComEd).

Overload 14: MISO: 1313/1591 MVA. ComEd SSTE: 1134 MVA (Violation for ComEd)

Overload 15: MISO: 1313/1591 MVA. ComEd SSTE: 1134 MVA (Violation for ComEd)

Overload 16: MISO: 1313/1591 MVA. ComEd SSTE: 1134 MVA (Violation for ComEd)

Overload 17: MISO: 1508 SE MVA. ComEd SSTE: 1483 MVA (Violation valid for ComEd & MISO).

Overload 18: MISO: 1508 SE MVA. ComEd SSTE: 1483 MVA (Violation valid for ComEd & MISO).

Overload 19: MISO: 1508 SE MVA. ComEd SSTE: 1483 MVA (Violation valid for ComEd & MISO).

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

To be determined.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

Overload Number	Type	Contingency Name	Affected Area	Facility Description	Bus		Circuit	Power Flow	Loading %		Rating Type	MV A	MW Contribution	Flowgate Appendix
					From	To			Initial	Final				
20	N-1	AEP_P1-2_#695A	MISO NIPS - CE	17STJOHN-ST JOHN ; T 345 kV line	25511 2	27088 6	1	DC	121.9 6	131.5 6	ER	1091	104.3	
21	N-1	AEP_P1-2_#695A	MISO NIPS - AEP	17STILLWELL-05DUMONT 345 kV line	25511 3	24321 9	1	AC	123.5 2	124.7	NR	1409	58.78	
22	N-1	COMED_P1-2_345-L94507_B-S	CE - CE	E FRANKFO; B-GOODINGS ;3B 345 kV line	27072 8	27076 6	1	AC	118.1 5	132.4 6	ER	1726	254.86	
23	Non	Non	CE - CE	E FRANKFO; B-GOODINGS ;3B 345 kV line	27072 8	27076 6	1	AC	119.7 1	129.3 9	NR	1334	134.19	
24	N-1	AEP_P1-2_#695A	CE - MISO NIPS	ST JOHN ; T-17GREEN_ACRE 345 kV line	27088 6	25510 4	1	DC	121.9 5	131.5 5	ER	1091	104.3	
25	N-1	AEP_P1-2_#695A	CE - MISO NIPS	CRETE EC ;BP-17STJOHN 345 kV line	27475 0	25511 2	1	AC	146.6 1	159.4 4	ER	1399	178.47	
26	Non	Non	CE - MISO NIPS	CRETE EC ;BP-17STJOHN 345 kV line	27475 0	25511 2	1	AC	91.08	106.4 1	NR	1091	167.32	
27	N-1	COMED_P1-2_345-L94507_B-S	CE - CE	CRETE EC ;BP-E FRANKFO; B 345 kV line	27475 0	27072 8	1	AC	125.7 7	150.4 6	ER	1399	356.45	
28	Non	Non	CE - CE	CRETE EC ;BP-E FRANKFO; B 345 kV line	27475 0	27072 8	1	AC	113.9 8	130.6 7	NR	1091	189.13	

Short Circuit

(Summary of impacted circuit breakers)

To be determined

Affected System Analysis & Mitigation

MISO Impacts:

Preliminary MISO impacts have been identified. Please refer to the MISO Affected System report for details. Final MISO Impacts to be determined in the Facilities Study phase.

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

To be determined during the Facilities Study.

Light Load Analysis – 2022

The Queue Project AE1-198 was evaluated as a 89 MW (Capacity 89 MW) withdrawal at Crete 345 kV substation in the ComEd area. Project AE1-198 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-198 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

Overload Number	Contingency		Affected Area	Facility Description	Bus		Circuit	Power Flow	Loading %		Rating Type	MVA	MW Contribution	Flowgate Appendix
	Type	Name			From	To			Initial	Final				
1	N-1	AEP_P1-2_#695A	CE - CE	E FRANKFO; B-CRETE EC ;BP 345 kV line	270728	274750	1	DC	104.74	107.95	ER	1399	44.97	9
2	LFFB	AEP_P4_#2978_05DUMONT 765_B	CE - CE	E FRANKFO; B-CRETE EC ;BP 345 kV line	270728	274750	1	AC	103.08	106.17		1399	45.06	
3	LFFB	COMED_P4_023-65-BT2-3__	CE - CE	E FRANKFO; B-CRETE EC ;BP 345 kV line	270728	274750	1	AC	102.86	105.97		1399	44.98	9

Note:

Violation 1: ComEd 1483 MVA SSTE (Violation valid)

Violation 2: ComEd 1483 MVA SSTE (Violation valid)

Violation 3: ComEd 1483 MVA SSTE (Not a Violation)

System Reinforcements

Summer Peak Load Flow Analysis Reinforcement

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None.

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

Facility	Upgrade Description	Cost	Cost Allocation	Upgrade Number
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17STJOHN-ST JOHN ; T 345 kV Ckt. 1	<u>MISO Reinforcement:</u> MISO Rating 1313/1591 MVA sufficient. No reinforcement required. <u>ComEd Reinforcement:</u> Project ID: n5833 Description: The upgrade will be to mitigate the sag on the line. Upon completion of this work the new line will be a minimum of 1091/1399/1483/1674 MVA (SN/SLTE/SSTE/SLD). PJM Network Upgrade N5833. Type: FAC Cost: \$3,800,000 Time Estimate: 30 Months Ratings: 1091/1399/1483/1674 MVA (SN/SLTE/SSTE/SLD) <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost</th></tr><tr><td>AE1-194</td><td>146.07</td><td>72.1 %</td><td>\$2,739,000</td></tr><tr><td>AE1-198</td><td>56.56</td><td>27.9 %</td><td>\$1,061,000</td></tr></table>	Queue	MW	Cost %	Cost	AE1-194	146.07	72.1 %	\$2,739,000	AE1-198	56.56	27.9 %	\$1,061,000	\$3,800,000	\$1,061,000	N5833
Queue	MW	Cost %	Cost													
AE1-194	146.07	72.1 %	\$2,739,000													
AE1-198	56.56	27.9 %	\$1,061,000													
17STILLWELL- 05DUMONT 345 kV Ckt. 1	<u>MISO Reinforcement:</u> Project ID: N/A Description: Upgrade Stillwell substation terminal equipment (upgrade substation conductor to bundled 954 ACSR and replace wave trap. Type: FAC Cost: \$1,500,000 Time Estimate: N/A Ratings: 1832 MVA SE Notes: AE1-198 is driver and is cost allocated 100%. <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost</th></tr><tr><td>AE1-198</td><td>56.56</td><td>100%</td><td>\$1,500,000</td></tr></table> <u>AEP Reinforcement:</u> Project ID: n4058 Description: Sag mitigation work will include the replacement of tower 20 with a custom steel pole, replacement of tower 24 with a custom H-frame and the removal of swing angle brackets on 2 structures. Cost estimate is \$1.613M. New SE rating will be 1718 MVA limited by a Dumont wave trap. Type: FAC Cost: \$1,613,000 Time Estimate: Projected in service date 6/1/2022 Ratings: AEP SE: 1718 MVA Project ID: n4790 Description: Upgrade Dumont 2500A wave trap. Type: FAC Cost: \$200,000 Time Estimate: N/A Ratings: AEP SE: 1740/1790 MVA (SN/SE) Notes: Since the cost of the upgrade is less than \$5M, based on PJM cost allocation criteria, AE1-198 currently does not receive cost allocation towards this upgrade. Project ID: n5769.1	Queue	MW	Cost %	Cost	AE1-198	56.56	100%	\$1,500,000	\$3,938,000	\$2,125,000	N4058 N4790 N5769.1				
Queue	MW	Cost %	Cost													
AE1-198	56.56	100%	\$1,500,000													

	Description: An Engineering study (\$25,000) will need to be conducted to determine if the Dumont CT Thermal Limit settings can be adjusted to mitigate the overload. New relay package will be required if the settings cannot be adjusted (\$600,000) Type: FAC Cost: 1. Cost Estimate for the study: \$25,000 2. Cost Estimate for new relay package: \$600,000 Time Estimate: Projected in service date 6/1/2022 Ratings: 1409/1868 MVA SN/SE <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost</th></tr><tr><td>AE1-198</td><td>56.56</td><td>100%</td><td>\$625,000</td></tr></table>	Queue	MW	Cost %	Cost	AE1-198	56.56	100%	\$625,000			
Queue	MW	Cost %	Cost									
AE1-198	56.56	100%	\$625,000									
GREENACRE; T-05OLIVE 345 kV Ckt. 1	Project ID: n5913 Description: Upgrade is a sag study will be required for the entire 40.64 miles of ACSR/PE ~ 1414 ~ 62/19 Conductor section 1 to determine if the line can be operated above its emergency rating 971 MVA. Estimated Cost: \$162,560. If deemed necessary to rebuild the entire 40.64 miles of the section of the line, Estimated Cost: \$81,280,000. New expected SE rating is 1318 MVA. PJM NUN N5913. Type: FAC Cost: \$162,560 Time Estimate: N/A Ratings: 1318 MVA Notes: 1. Since the cost of the upgrade is less than \$5M, based on PJM cost allocation criteria, AE1-198 currently does not receive cost allocation towards this upgrade. 2. As changes to the PJM queue process occur (such as prior queued projects withdrawing from the queue, reducing in size, etc.) AE1-198 could receive cost allocation. 3. Although Queue Project AE1-198 may not presently have cost responsibility for this upgrade, Queue Project AE1-198 may need this upgrade in-service to be deliverable to the PJM system. 4. If Queue Project AE1-198 comes into service prior to completion of the upgrade, Queue Project AE1-198 will need an interim study.	\$162,560	\$0	N5913								
17ST JOHN ; T - 17GREEN_ACRE 345 kV line	ComEd ComEd SSTE rating is 1134 MVA. ComEd Reinforcement: Project ID: n5834 Description: The upgrade is to mitigate the sag on the line. A preliminary estimate for this upgrade is \$3.8 M with a preliminary construction timeline of 30 months. The estimate provided does not include potential tower upgrades. The cost for this potential work will not be identified until the Facilities Study phase. Upon completion of the upgrade the new ratings will be 1091/1399/1483/1674 MVA (SN/SLTE/SSTE/SLD). Type: FAC Cost: \$3,800,000 Time Estimate: 30 Months Ratings: 1091/1399/1483/1674 MVA (SN/SLTE/SSTE/SLD)	\$3,800,000	\$1,583,017	n5834								

	Notes: Portion of this facility was purchased and is now owned by NEXTERA. NEXTERA will evaluate this violation in the Facility Study phase. <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost</th></tr><tr><td>AE1-194</td><td>146.07</td><td>58%</td><td>\$2,216,983</td></tr><tr><td>AE1-198</td><td>104.3</td><td>42%</td><td>\$1,583,017</td></tr></table> MISO MISO-end ratings (1313/1591 MVA -SN/SE) are sufficient.	Queue	MW	Cost %	Cost	AE1-194	146.07	58%	\$2,216,983	AE1-198	104.3	42%	\$1,583,017																							
Queue	MW	Cost %	Cost																																	
AE1-194	146.07	58%	\$2,216,983																																	
AE1-198	104.3	42%	\$1,583,017																																	
Crete – St John 345 kV line Ckt 1	ComEd ComEd SSTE rating is 1483 MVA. ComEd Reinforcement: Project ID: n5253 Description: The upgrade will be to Reconductor the line. A preliminary estimate for this upgrade is \$14.9M with a preliminary construction timeline of 30-36 months. The estimate provided does not consider potential tower upgrades. This cost component will be determined during the Facilities Study phase. Upon completion of this upgrade, the new will be a minimum of 1334/1726/1837/2084 MVA (SN/SLTE/SSTE/SLD). Type: FAC Cost: \$14,900,000 Time Estimate: 30-36 Months Ratings: 1334/1726/1837/2084 MVA (SN/SLTE/SSTE/SLD) The cost allocation is as follows: <table><tr><th>Queue</th><th>MW contribution</th><th>Cost %</th><th>cost (\$14.9 M)</th></tr><tr><td>AE1-193</td><td>251.82</td><td>27.02%</td><td>\$4,025,700</td></tr><tr><td>AE1-194</td><td>249.95</td><td>27.02%</td><td>\$4,025,700</td></tr><tr><td>AE1-195</td><td>249.95</td><td>26.82%</td><td>\$3,995,767</td></tr><tr><td>AE1-198</td><td>178.46</td><td>19.5%</td><td>\$2,852,909</td></tr></table> Project ID: n6629 Description: ComEd upgrade: Replace a 345kV circuit breaker and associated equipment at Crete. A preliminary estimate for this upgrade is \$6 M with a preliminary construction timeline of 30 months. Upon completion of this upgrade the new ratings will be 1754/2246/2297/2488 MVA (SN/SLTE/SSTE/SLD). PJM Network Upgrade N6629 Type: FAC Cost: \$6,000,000 Time Estimate: 30 Months Ratings: 1754/2246/2297/2488 MVA (SN/SLTE/SSTE/SLD) The cost allocation is as follows: <table><tr><th>Queue</th><th>MW contribution</th><th>Cost %</th><th>cost (\$14.9 M)</th></tr><tr><td>AE1-195</td><td>248.03</td><td>58.16%</td><td>\$3,489,000</td></tr><tr><td>AE1-198</td><td>178.46</td><td>41.84%</td><td>\$2,511,000</td></tr></table>	Queue	MW contribution	Cost %	cost (\$14.9 M)	AE1-193	251.82	27.02%	\$4,025,700	AE1-194	249.95	27.02%	\$4,025,700	AE1-195	249.95	26.82%	\$3,995,767	AE1-198	178.46	19.5%	\$2,852,909	Queue	MW contribution	Cost %	cost (\$14.9 M)	AE1-195	248.03	58.16%	\$3,489,000	AE1-198	178.46	41.84%	\$2,511,000	\$22,900,000	\$6,461,701	N5253 N6629 N6603
Queue	MW contribution	Cost %	cost (\$14.9 M)																																	
AE1-193	251.82	27.02%	\$4,025,700																																	
AE1-194	249.95	27.02%	\$4,025,700																																	
AE1-195	249.95	26.82%	\$3,995,767																																	
AE1-198	178.46	19.5%	\$2,852,909																																	
Queue	MW contribution	Cost %	cost (\$14.9 M)																																	
AE1-195	248.03	58.16%	\$3,489,000																																	
AE1-198	178.46	41.84%	\$2,511,000																																	

	Notes: This facility was purchased and is now owned by NEXTERA. NEXTERA will evaluate this violation in the Facility Study phase. MISO MISO end ratings are 1206/1508 MVA (SN/SE) MISO Reinforcement: Project ID: N/A Description: MISO-End upgrade #1: Upgrade St John substation conductor drop and switch. \$1M cost estimate. New MISO end SE rating will be 1900 MVA. Type: FAC Cost: \$1,000,000 Time Estimate: N/A Months Ratings: 1900 MVA SE <table><thead><tr><th>Queue</th><th>MW contribution</th><th>Cost %</th><th>cost (\$3M)</th></tr></thead><tbody><tr><td>AE1-193</td><td>251.82</td><td>37.02%</td><td>\$370,177</td></tr><tr><td>AE1-194</td><td>249.95</td><td>36.74%</td><td>\$367,428</td></tr><tr><td>AE1-198</td><td>178.46</td><td>26.24%</td><td>\$262,396</td></tr></tbody></table> Project ID: N/A Description: MISO-end upgrade #2: Upgrade switch to 4000 A and upgrade substation conductor drop to bundled 1590 AL. \$2M. New MISO-end ratings to be 1961/2390 MVA SN/SE. Type: FAC Cost: \$2,000,000 Time Estimate: N/A Months Ratings: 1961/2390 MVA SN/SE <table><thead><tr><th>Queue</th><th>MW contribution</th><th>Cost %</th><th>cost (\$3M)</th></tr></thead><tbody><tr><td>AE1-195</td><td>249</td><td>58.25%</td><td>\$1,164,990</td></tr><tr><td>AE1-198</td><td>178.5</td><td>41.75%</td><td>\$835,010</td></tr></tbody></table>	Queue	MW contribution	Cost %	cost (\$3M)	AE1-193	251.82	37.02%	\$370,177	AE1-194	249.95	36.74%	\$367,428	AE1-198	178.46	26.24%	\$262,396	Queue	MW contribution	Cost %	cost (\$3M)	AE1-195	249	58.25%	\$1,164,990	AE1-198	178.5	41.75%	\$835,010			
Queue	MW contribution	Cost %	cost (\$3M)																													
AE1-193	251.82	37.02%	\$370,177																													
AE1-194	249.95	36.74%	\$367,428																													
AE1-198	178.46	26.24%	\$262,396																													
Queue	MW contribution	Cost %	cost (\$3M)																													
AE1-195	249	58.25%	\$1,164,990																													
AE1-198	178.5	41.75%	\$835,010																													
	Total Cost	\$33,600,560	\$11,230,718																													

Short Circuit System Reinforcement

(Summary form of Cost allocation for breakers will be inserted here if any)

To be determined

Contingencies (Summer Peak Analysis)

Contingency Name	Description
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AEP_P1-2_#695A	CONTINGENCY 'AEP_P1-2_#695A' OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
AEP_P4_#2978_05DUMONT 765_B	CONTINGENCY 'AEP_P4_#2978_05DUMONT 765_B' OPEN BRANCH FROM BUS 243206 TO BUS 243207 CKT 1 / 243206 05DUMONT 765 243207 05GRNTWN 765 1 OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
COMED_P1-2_345-L94507_B-S	CONTINGENCY 'COMED_P1-2_345-L94507_B-S' TRIP BRANCH FROM BUS 274750 TO BUS 255112 CKT 1 / CRETE;BP 345 17STJOHN 345 END
COMED_P2-2_116_GG-345R__2_FSA	CONTINGENCY 'COMED_P2-2_116_GG-345R__2_FSA' DISCONNECT BUS 270769 / GOODI;2R 345 DISCONNECT BUS 918050 / AA1-018 TAP /ADDED FOR NO FSA CASE END
COMED_P2-2_116_GG-345R__2_NO_FSA	CONTINGENCY 'COMED_P2-2_116_GG-345R__2_NO_FSA' DISCONNECT BUS 270769 / GOODI;2R 345 END
COMED_P4_023-65-BT2-3__	CONTINGENCY 'COMED_P4_023-65-BT2-3__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 270607 TO BUS 270630 CKT 1 / COLLI; 765 PLANO; 765 END
COMED_P4_112-65-BT3-4__	CONTINGENCY 'COMED_P4_112-65-BT3-4__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 275232 TO BUS 270644 CKT 1 / WILTO;3M 345 WILTO; 765 TRIP BRANCH FROM BUS 275232 TO BUS 270926 CKT 1 / WILTO;3M 345 WILTO; B 345 TRIP BRANCH FROM BUS 275232 TO BUS 275332 CKT 1 / WILTO;3M 345 WILTO;3C 33

	END
COMED_P4_112-65-BT4-5__	CONTINGENCY 'COMED_P4_112-65-BT4-5__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 275233 TO BUS 270644 CKT 1 / WILTO;4M 345 WILTO; 765 TRIP BRANCH FROM BUS 275233 TO BUS 270927 CKT 1 / WILTO;4M 345 WILTO; R 345 TRIP BRANCH FROM BUS 275233 TO BUS 275333 CKT 1 / WILTO;4M 345 WILTO;4C 33 END
COMED_P4_116-45-L11614_	CONTINGENCY 'COMED_P4_116-45-L11614_' TRIP BRANCH FROM BUS 270667 TO BUS 270665 CKT 1 / B ISL;RT 345 B ISL; R 345 TRIP BRANCH FROM BUS 270667 TO BUS 270927 CKT 1 / B ISL;RT 345 WILTO; R 345 TRIP BRANCH FROM BUS 270769 TO BUS 270667 CKT 1 / GOODI;2R 345 B ISL;RT 345 DISCONNECT BUS 270769 / GOODI;2R 345 END
COMED_P4_116-45-L9801_FSA	CONTINGENCY 'COMED_P4_116-45-L9801_FSA' TRIP BRANCH FROM BUS 918050 TO BUS 270769 CKT 1 / AA1-018 GOODINGS ;2R 345 DISCONNECT BUS 270769 / GOODI;2R 345 END
COMED_P7_345-L94507_B-S+_345-L97008_R-S	CONTINGENCY 'COMED_P7_345-L94507_B-S+_345-L97008_R-S' TRIP BRANCH FROM BUS 274750 TO BUS 255112 CKT 1 / CRETE;BP 345 17STJOHN 345 TRIP BRANCH FROM BUS 274804 TO BUS 243229 CKT 1 / UPNOR;RP 345 05OLIVE 345 END

Light Load Peak Load Flow Analysis Reinforcement

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None.

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

Facility	Upgrade Description	Cost	Cost Allocation	Upgrade Number								
E FRANKFO; B-CRETE EC ;BP 345 kV Ckt 1	Project ID: n5917 Description: L6607 Ratings MVA: 1483 SSTE/1925 ALDR. The post contingency flow for this event exceeds the rating therefore an upgrade is required. The upgrade will be to Reconductor the line. A preliminary estimate is \$10.3 M with an estimated construction timeline of 30 months. Upon completion of the upgrades the rating swill be 1334/1726/1837/2084 MVA (SN/SLTE/SSTE/SLD). Type: FAC Cost: \$10,300,000 Time Estimate: 30 Months Ratings:1334/1726/1837/2084 MVA (SN/SLTE/SSTE/SLD) <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost</th></tr><tr><td>AE1-198</td><td>27</td><td>100%</td><td>\$10,300,000</td></tr></table>	Queue	MW	Cost %	Cost	AE1-198	27	100%	\$10,300,000	\$10,300,000	\$10,300,000	N5917
Queue	MW	Cost %	Cost									
AE1-198	27	100%	\$10,300,000									
	Total Cost	\$10,300,000	\$10,300,000									

Short Circuit System Reinforcement

(Summary form of Cost allocation for breakers will be inserted here if any)

To be determined

Contingencies (Light Load)

Option 1	
Contingency Name	Description
AEP_P1-2_#695A	CONTINGENCY 'AEP_P1-2_#695A' OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
AEP_P4_#2978_05DUMONT 765_B	CONTINGENCY 'AEP_P4_#2978_05DUMONT 765_B' OPEN BRANCH FROM BUS 243206 TO BUS 243207 CKT 1 / 243206 05DUMONT 765 243207 05GRNTWN 765 1 OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
COMED_P4_023-65-BT2-3__	CONTINGENCY 'COMED_P4_023-65-BT2-3__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 270607 TO BUS 270630 CKT 1 / COLLI; 765 PLANO; 765 END

Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the Appendices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the Appendices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators.

It should be noted the project/generator MW contributions presented in the body of the report and appendices sections are full contributions, whereas the loading percentages reported in the body of the report, take into consideration the commercial probability of each project as well as the ramping impact of "Adder" contributions.

Appendix 1

(CE - CE) The E FRANKFO; B-GOODINGS ;3B 345 kV line (from bus 270728 to bus 270766 ckt 1) loads from 97.86% to 109.74% (AC power flow) of its load dump rating (2084 MVA) for the tower line contingency outage of 'COMED_P7_345-L94507_B-S+_345-L97008_R-S'. This project contributes approximately 255.42 MW to the thermal violation.

CONTINGENCY 'COMED_P7_345-L94507_B-S+_345-L97008_R-S'

TRIP BRANCH FROM BUS 274750 TO BUS 255112 CKT 1 / CRETE;BP 345
17STJOHN 345

TRIP BRANCH FROM BUS 274804 TO BUS 243229 CKT 1 / UPNOR;RP 345
05OLIVE 345

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
939351	AE1-166 C O1	8.03
939352	AE1-166 E O1	7.42
939631	AE1-193 C O1	46.5
939632	AE1-193 E O1	311.23
939641	AE1-194 C	46.5
939642	AE1-194 E	311.23
939651	AE1-195 C	46.5
939652	AE1-195 E	311.23
939681	AE1-198 C O1	138.08
939682	AE1-198 E O1	117.34
LTF	BLUEG	0.84
274654	BRAIDWOOD;1U	30.75
274655	BRAIDWOOD;2U	29.18
LTF	CALDERWOOD	0.15
LTF	CANNELTON	0.06
LTF	CARR	0.07
LTF	CATAWBA	0.08
LTF	CHEOAH	0.13
LTF	COFFEEN	0.03
LTF	COTTONWOOD	0.91
274751	CRETE EC ;1U	9.48
274752	CRETE EC ;2U	9.48
274753	CRETE EC ;3U	9.48
274754	CRETE EC ;4U	9.48
LTF	DUCKCREEK	2.06
LTF	EDWARDS	1.
LTF	FARMERCITY	0.36
LTF	G-007	0.21
LTF	GIBSON	0.03

<i>LTF</i>	<i>HAMLET</i>	<i>0.14</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.15</i>
<i>LTF</i>	<i>O-066</i>	<i>1.35</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>0.9</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>0.06</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.06</i>
<i>LTF</i>	<i>TATANKA</i>	<i>0.87</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.09</i>
<i>LTF</i>	<i>TVA</i>	<i>0.54</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.31</i>

Appendix 2

(MISO NIPS - CE) The 17GREEN_ACRE-GREENACRE; T 345 kV line (from bus 255104 to bus 270771 ckt 1) loads from 100.16% to 100.54% (AC power flow) of its emergency rating (1091 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#2978_05DUMONT 765_B'. This project contributes approximately 59.94 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#2978_05DUMONT 765_B'

OPEN BRANCH FROM BUS 243206 TO BUS 243207 CKT 1 / 243206
05DUMONT 765 243207 05GRNTWN 765 1

OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206
05DUMONT 765 270644 WILTON ; 765 1

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	1.59
932891	AC2-115 2	1.59
932921	AC2-116	0.56
933411	AC2-154 C	1.69
933412	AC2-154 E	2.76
933911	AD1-013 C	1.23
933912	AD1-013 E	1.96
933931	AD1-016 C	0.62
933932	AD1-016 E	1.02
934051	AD1-031 C O1	1.88
934052	AD1-031 E O1	3.07
934101	AD1-039 1	4.69
934111	AD1-039 2	4.9
934431	AD1-067 C	0.09
934432	AD1-067 E	0.37
934701	AD1-098 C O1	4.6
934702	AD1-098 E O1	3.35
934721	AD1-100 C	13.01
934722	AD1-100 E	60.71
934871	AD1-116 C	0.62
934872	AD1-116 E	1.01
934971	AD1-129 C	0.61
934972	AD1-129 E	0.4
935001	AD1-133 C O1	13.81
935002	AD1-133 E O1	9.21
936291	AD2-038 C O1	1.55
936292	AD2-038 E O1	10.38

936371	AD2-047 C O1	3.03
936372	AD2-047 E O1	14.78
936461	AD2-060	1.78
936511	AD2-066 C O1	5.6
936512	AD2-066 E O1	3.73
936791	AD2-102 C	9.47
936792	AD2-102 E	6.31
937001	AD2-134 C	1.73
937002	AD2-134 E	6.9
937311	AD2-172 C	1.65
937312	AD2-172 E	2.27
937401	AD2-194 1	5.27
937411	AD2-194 2	5.27
937531	AD2-214 C	2.94
937532	AD2-214 E	1.96
938511	AE1-070 1	6.19
938521	AE1-070 2	5.66
938851	AE1-113 C O1	5.39
938852	AE1-113 E O1	19.1
938861	AE1-114 C O1	2.63
938862	AE1-114 E O1	8.96
939051	AE1-134 1	0.91
939061	AE1-134 2	0.91
939321	AE1-163 C O1	3.9
939322	AE1-163 E O1	23.94
939351	AE1-166 C O1	6.87
939352	AE1-166 E O1	6.34
939401	AE1-172 C O1	3.5
939402	AE1-172 E O1	16.4
939631	AE1-193 C O1	10.91
939632	AE1-193 E O1	73.04
939641	AE1-194 C	10.91
939642	AE1-194 E	73.04
939651	AE1-195 C	10.91
939652	AE1-195 E	73.04
939681	AE1-198 C O1	32.41
939682	AE1-198 E O1	27.54
939741	AE1-205 C O1	5.79
939742	AE1-205 E O1	8.
940101	AE1-252 C O1	7.02
940102	AE1-252 E O1	4.68
LTF	BLUEG	2.63
294401	BSHIL;1U E	5.66
294410	BSHIL;2U E	5.66
LTF	CARR	0.48

LTF	CATAWBA	0.15
274890	CAYUG;1U E	8.97
274891	CAYUG;2U E	8.97
LTF	CBM-S1	1.49
LTF	CBM-W1	20.35
LTF	CBM-W2	39.71
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
LTF	CIN	0.23
274751	CRETE EC ;1U	2.22
274752	CRETE EC ;2U	2.22
274753	CRETE EC ;3U	2.22
274754	CRETE EC ;4U	2.22
274859	EASYR;U1 E	7.34
274860	EASYR;U2 E	7.34
LTF	G-007	1.33
290051	GSG-6; E	6.98
LTF	HAMLET	0.3
953871	J847	2.7
275149	KEMPTON ;1E	12.46
990901	L-005 E	8.29
290108	LEEDK;1U E	16.23
LTF	MEC	25.26
293061	N-015 E	10.29
293516	O-009 E1	6.08
293517	O-009 E2	3.09
293518	O-009 E3	3.4
293715	O-029 E	6.5
293716	O-029 E	3.56
293717	O-029 E	3.28
293771	O-035 E	4.23
LTF	O-066	8.55
293644	O22 E1	7.28
293645	O22 E2	14.13
290021	O50 E	13.06
294392	P-010 E	13.07
294763	P-046 E	6.27
274888	PILOT HIL;1E	12.46
270859	PWR VTR EC;R	8.1
LTF	RENSSELAER	0.38
274722	S-055 E	7.56
295111	SUBLETTE E	1.82
LTF	TRIMBLE	0.32
LTF	WEC	5.35
295109	WESTBROOK E	3.74

910542	X3-005 E	0.52
915011	Y3-013 1	2.52
915021	Y3-013 2	2.52
915031	Y3-013 3	2.52
916211	Z1-072 E	3.2
916221	Z1-073 E	3.6
916502	Z1-106 E1	0.85
916504	Z1-106 E2	0.85
916512	Z1-107 E	1.72
916522	Z1-108 E	1.67
917502	Z2-087 E	11.99
918052	AA1-018 E	10.68
919221	AA1-146	11.73
919581	AA2-030	11.73
920272	AA2-123 E	1.64
930481	AB1-089	44.01
930501	AB1-091 O1	49.4
930741	AB1-122 1O1	47.85
930751	AB1-122 2O1	50.02
924041	AB2-047 C O1	2.24
924042	AB2-047 E O1	14.99
924471	AB2-096	28.36
925302	AB2-191 E	0.92
925581	AC1-033 C	0.93
925582	AC1-033 E	6.21
926311	AC1-109 1	1.28
926321	AC1-109 2	1.28
926331	AC1-110 1	1.27
926341	AC1-110 2	1.27
926351	AC1-111 1	0.51
926361	AC1-111 2	0.51
926371	AC1-111 3	0.51
926381	AC1-111 4	0.51
926391	AC1-111 5	0.51
926401	AC1-111 6	0.51
927511	AC1-113 1	0.8
927521	AC1-113 2	0.8
926431	AC1-114	1.59
927451	AC1-142A 1	2.85
927461	AC1-142A 2	2.85
926821	AC1-168 C O1	0.76
926822	AC1-168 E O1	5.11
927091	AC1-204 1	49.02
927101	AC1-204 2	48.98
927201	AC1-214 C O1	1.36

927202	<i>AC1-214 E O1</i>	4.32
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Appendix 3

(MISO NIPS - CE) The 17STJOHN-ST JOHN ; T 345 kV line (from bus 255112 to bus 270886 ckt 1) loads from 122.97% to 132.32% (AC power flow) of its emergency rating (1091 MVA) for the line fault with failed breaker contingency outage of 'COMED_P4_023-65-BT2-3__'. This project contributes approximately 104.3 MW to the thermal violation.

CONTINGENCY 'COMED_P4_023-65-BT2-3__'

TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765
05DUMONT 765

TRIP BRANCH FROM BUS 270607 TO BUS 270630 CKT 1 / COLLI; 765
PLANO; 765

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	1.78
932891	AC2-115 2	1.78
932921	AC2-116	0.62
933411	AC2-154 C	1.75
933412	AC2-154 E	2.85
933911	AD1-013 C	1.38
933912	AD1-013 E	2.2
933931	AD1-016 C	0.7
933932	AD1-016 E	1.14
934101	AD1-039 1	5.22
934111	AD1-039 2	5.55
934431	AD1-067 C	0.1
934432	AD1-067 E	0.41
934701	AD1-098 C O1	5.14
934702	AD1-098 E O1	3.76
934721	AD1-100 C	14.35
934722	AD1-100 E	66.95
934871	AD1-116 C	0.68
934872	AD1-116 E	1.11
934971	AD1-129 C	0.68
934972	AD1-129 E	0.45
935001	AD1-133 C O1	15.25
935002	AD1-133 E O1	10.17
936291	AD2-038 C O1	1.72
936292	AD2-038 E O1	11.52
936371	AD2-047 C O1	3.13
936372	AD2-047 E O1	15.27
936461	AD2-060	1.84

936511	AD2-066 C O1	6.23
936512	AD2-066 E O1	4.15
936791	AD2-102 C	10.6
936792	AD2-102 E	7.07
937001	AD2-134 C	1.94
937002	AD2-134 E	7.73
937311	AD2-172 C	1.84
937312	AD2-172 E	2.54
937401	AD2-194 1	5.97
937411	AD2-194 2	5.96
938511	AE1-070 1	7.01
938521	AE1-070 2	6.4
938851	AE1-113 C O1	6.08
938852	AE1-113 E O1	21.55
938861	AE1-114 C O1	2.94
938862	AE1-114 E O1	10.02
939051	AE1-134 1	1.02
939061	AE1-134 2	1.02
939321	AE1-163 C O1	4.32
939322	AE1-163 E O1	26.57
939351	AE1-166 C O1	7.64
939352	AE1-166 E O1	7.05
939401	AE1-172 C O1	3.83
939402	AE1-172 E O1	17.91
939631	AE1-193 C O1	18.99
939632	AE1-193 E O1	127.09
939641	AE1-194 C	18.99
939642	AE1-194 E	127.09
939651	AE1-195 C	18.99
939652	AE1-195 E	127.09
939681	AE1-198 C O1	56.38
939682	AE1-198 E O1	47.91
940101	AE1-252 C O1	7.67
940102	AE1-252 E O1	5.12
LTF	BLUEG	4.79
274654	BRAIDWOOD;1U	22.
274655	BRAIDWOOD;2U	21.06
LTF	CALDERWOOD	0.06
LTF	CANNELTON	0.09
LTF	CARR	0.54
LTF	CATAWBA	0.21
274890	CAYUG;1U E	8.79
274891	CAYUG;2U E	8.79
LTF	CBM-S1	0.91
LTF	CBM-W1	20.43

LTF	CBM-W2	38.51
LTF	CHEOAH	0.06
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
274751	CRETE EC ;1U	3.87
274752	CRETE EC ;2U	3.87
274753	CRETE EC ;3U	3.87
274754	CRETE EC ;4U	3.87
274859	EASYR;U1 E	8.22
274860	EASYR;U2 E	8.22
LTF	G-007	1.52
LTF	GIBSON	0.05
290051	GSG-6; E	7.82
LTF	HAMLET	0.4
275149	KEMPTON ;1E	12.88
274704	KENDALL ;1C	3.25
274705	KENDALL ;1S	2.17
274706	KENDALL ;2C	3.25
274707	KENDALL ;2S	2.17
274661	LASCO STA;2U	20.32
290108	LEEDK;1U E	18.19
LTF	MEC	27.81
293061	N-015 E	11.68
293516	O-009 E1	6.79
293517	O-009 E2	3.45
293518	O-009 E3	3.8
293715	O-029 E	7.26
293716	O-029 E	3.98
293717	O-029 E	3.66
LTF	O-066	9.77
293644	O22 E1	8.54
293645	O22 E2	16.59
290021	O50 E	14.73
294392	P-010 E	14.83
294763	P-046 E	7.02
274888	PILOT HIL;1E	12.88
270859	PWR VTR EC;R	9.09
LTF	RENSSELAER	0.43
274722	S-055 E	8.49
295111	SUBLETTE E	2.04
274861	TOP CROP ;1U	0.38
274862	TOP CROP ;2U	0.73
LTF	TRIMBLE	0.56
LTF	WEC	6.
295109	WESTBROOK E	4.19

274687	WILL CNTY;4U	9.27
915011	Y3-013 1	2.83
915021	Y3-013 2	2.83
915031	Y3-013 3	2.83
916221	Z1-073 E	4.03
916502	Z1-106 E1	0.95
916504	Z1-106 E2	0.95
916512	Z1-107 E	1.86
916522	Z1-108 E	1.87
918052	AA1-018 E	11.7
919221	AA1-146	13.12
919581	AA2-030	13.12
920272	AA2-123 E	1.84
930481	AB1-089	49.32
930501	AB1-091 O1	50.89
930741	AB1-122 1O1	53.27
930751	AB1-122 2O1	56.61
924471	AB2-096	31.8
925302	AB2-191 E	1.04
926311	AC1-109 1	1.43
926321	AC1-109 2	1.43
926331	AC1-110 1	1.43
926341	AC1-110 2	1.43
926351	AC1-111 1	0.57
926361	AC1-111 2	0.57
926371	AC1-111 3	0.57
926381	AC1-111 4	0.57
926391	AC1-111 5	0.57
926401	AC1-111 6	0.57
927511	AC1-113 1	0.89
927521	AC1-113 2	0.89
926431	AC1-114	1.78
927451	AC1-142A 1	3.22
927461	AC1-142A 2	3.22
926821	AC1-168 C O1	0.85
926822	AC1-168 E O1	5.72
927091	AC1-204 1	55.48
927101	AC1-204 2	55.4

Appendix 4

(MISO NIPS - AEP) The 17STILLWELL-05DUMONT 345 kV line (from bus 255113 to bus 243219 ckt 1) loads from 126.68% to 127.81% (AC power flow) of its emergency rating (1409 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#2978_05DUMONT 765_B'. This project contributes approximately 56.56 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#2978_05DUMONT 765_B'

OPEN BRANCH FROM BUS 243206 TO BUS 243207 CKT 1 / 243206
05DUMONT 765 243207 05GRNTWN 765 1

OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206
05DUMONT 765 270644 WILTON ; 765 1

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	2.76
932891	AC2-115 2	2.76
932921	AC2-116	0.96
932931	AC2-117	5.85
933411	AC2-154 C	3.04
933412	AC2-154 E	4.96
933911	AD1-013 C	2.13
933912	AD1-013 E	3.4
933931	AD1-016 C	1.07
933932	AD1-016 E	1.75
934051	AD1-031 C O1	3.3
934052	AD1-031 E O1	5.39
934101	AD1-039 1	8.13
934111	AD1-039 2	8.37
934431	AD1-067 C	0.15
934432	AD1-067 E	0.64
934701	AD1-098 C O1	7.96
934702	AD1-098 E O1	5.81
934721	AD1-100 C	22.58
934722	AD1-100 E	105.36
934871	AD1-116 C	1.1
934872	AD1-116 E	1.79
934971	AD1-129 C	1.05
934972	AD1-129 E	0.7
935001	AD1-133 C O1	24.22
935002	AD1-133 E O1	16.14
936291	AD2-038 C O1	2.71

936292	AD2-038 E O1	18.14
936371	AD2-047 C O1	5.44
936372	AD2-047 E O1	26.58
936461	AD2-060	3.2
936511	AD2-066 C O1	9.74
936512	AD2-066 E O1	6.49
936791	AD2-102 C	16.39
936792	AD2-102 E	10.93
937001	AD2-134 C	3.
937002	AD2-134 E	11.96
937311	AD2-172 C	2.85
937312	AD2-172 E	3.94
937401	AD2-194 1	9.
937411	AD2-194 2	9.
937531	AD2-214 C	5.12
937532	AD2-214 E	3.42
938511	AE1-070 1	10.58
938521	AE1-070 2	9.68
938851	AE1-113 C O1	9.26
938852	AE1-113 E O1	32.83
938861	AE1-114 C O1	4.56
938862	AE1-114 E O1	15.54
939051	AE1-134 1	1.59
939061	AE1-134 2	1.59
939321	AE1-163 C O1	6.81
939322	AE1-163 E O1	41.83
939351	AE1-166 C O1	11.86
939352	AE1-166 E O1	10.95
939401	AE1-172 C O1	6.16
939402	AE1-172 E O1	28.85
939631	AE1-193 C O1	10.3
939632	AE1-193 E O1	68.91
939641	AE1-194 C	10.3
939642	AE1-194 E	68.91
939651	AE1-195 C	10.3
939652	AE1-195 E	68.91
939681	AE1-198 C O1	30.58
939682	AE1-198 E O1	25.98
939741	AE1-205 C O1	10.32
939742	AE1-205 E O1	14.25
940101	AE1-252 C O1	12.36
940102	AE1-252 E O1	8.24
LTF	BLUEG	0.64
294401	BSHIL;1U E	9.93
294410	BSHIL;2U E	9.93

LTF	CARR	0.87
LTF	CATAWBA	0.19
274890	CAYUG;1U E	15.86
274891	CAYUG;2U E	15.86
LTF	CBM-S1	4.21
LTF	CBM-W1	36.52
LTF	CBM-W2	84.42
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
LTF	CIN	3.51
274859	EASYR;U1 E	12.75
274860	EASYR;U2 E	12.75
LTF	G-007	2.42
290051	GSG-6; E	12.09
LTF	HAMLET	0.43
LTF	IPL	1.25
954751	J351	165.42
275149	KEMPTON ;1E	22.42
990901	L-005 E	14.6
290108	LEEDK;1U E	28.09
LTF	MEC	45.08
293061	N-015 E	17.64
293516	O-009 E1	10.59
293517	O-009 E2	5.38
293518	O-009 E3	5.92
293715	O-029 E	11.32
293716	O-029 E	6.21
293717	O-029 E	5.71
293771	O-035 E	7.42
LTF	O-066	15.53
293644	O22 E1	12.02
293645	O22 E2	23.33
290021	O50 E	22.45
294392	P-010 E	22.4
294763	P-046 E	10.86
274888	PILOT HIL;1E	22.42
270859	PWR VTR EC;R	14.01
LTF	RENSSELAER	0.69
274724	RIVER EC ;11	5.49
274722	S-055 E	13.02
274795	SE CHICAG;2U	1.29
274788	SE CHICAG;5U	1.31
274789	SE CHICAG;6U	1.31
274790	SE CHICAG;7U	1.31
274791	SE CHICAG;8U	1.31

295111	SUBLETTE E	3.15
LTF	TRIMBLE	0.12
LTF	WEC	9.25
295109	WESTBROOK E	6.48
910542	X3-005 E	1.
915011	Y3-013 1	4.34
915021	Y3-013 2	4.34
915031	Y3-013 3	4.34
916211	Z1-072 E	5.61
916221	Z1-073 E	6.24
916502	Z1-106 E1	1.46
916504	Z1-106 E2	1.46
916512	Z1-107 E	3.05
916522	Z1-108 E	2.88
917502	Z2-087 E	21.38
918052	AA1-018 E	18.85
919221	AA1-146	20.41
919581	AA2-030	20.41
920272	AA2-123 E	2.83
930481	AB1-089	76.13
930501	AB1-091 O1	88.69
930741	AB1-122 1O1	82.91
930751	AB1-122 2O1	85.44
924041	AB2-047 C O1	3.99
924042	AB2-047 E O1	26.73
924471	AB2-096	49.02
925302	AB2-191 E	1.6
925581	AC1-033 C	1.63
925582	AC1-033 E	10.89
926311	AC1-109 1	2.2
926321	AC1-109 2	2.2
926331	AC1-110 1	2.19
926341	AC1-110 2	2.19
926351	AC1-111 1	0.88
926361	AC1-111 2	0.88
926371	AC1-111 3	0.88
926381	AC1-111 4	0.88
926391	AC1-111 5	0.88
926401	AC1-111 6	0.88
927511	AC1-113 1	1.38
927521	AC1-113 2	1.38
926431	AC1-114	2.76
927451	AC1-142A 1	4.86
927461	AC1-142A 2	4.86
926821	AC1-168 C O1	1.33

926822	<i>AC1-168 E O1</i>	<i>8.91</i>
927091	<i>AC1-204 1</i>	<i>83.73</i>
927101	<i>AC1-204 2</i>	<i>83.7</i>
927201	<i>AC1-214 C O1</i>	<i>2.38</i>
927202	<i>AC1-214 E O1</i>	<i>7.57</i>

Appendix 5

(CE - CE) The GOODINGS ;4B-GOODINGS ;3B 345 kV line (from bus 270770 to bus 270766 ckt 1) loads from 104.82% to 106.25% (**DC power flow**) of its load dump rating (1802 MVA) for the bus fault outage of 'COMED_P2-2_116_GG-345R__2_FSA'. This project contributes approximately 25.4 MW to the thermal violation.

CONTINGENCY 'COMED_P2-2_116_GG-345R__2_FSA'

DISCONNECT BUS 270769

/ GOODI;2R 345

DISCONNECT BUS 918050

/ AA1-018 TAP /ADDED FOR NO

FSA CASE

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	1.6
932891	AC2-115 2	1.6
932921	AC2-116	0.56
933931	AD1-016 C	0.61
933932	AD1-016 E	1.
934051	AD1-031 C O1	2.21
934052	AD1-031 E O1	3.6
934101	AD1-039 1	6.38
934111	AD1-039 2	9.41
934971	AD1-129 C	0.6
934972	AD1-129 E	0.4
935001	AD1-133 C O1	14.42
935002	AD1-133 E O1	9.61
936291	AD2-038 C O1	2.99
936292	AD2-038 E O1	19.99
936511	AD2-066 C O1	6.51
936512	AD2-066 E O1	4.34
936791	AD2-102 C	9.6
936792	AD2-102 E	6.4
937311	AD2-172 C	1.68
937312	AD2-172 E	2.33
937401	AD2-194 1	12.43
937411	AD2-194 2	12.36
937531	AD2-214 C	3.29
937532	AD2-214 E	2.19
938511	AE1-070 1	14.6
938521	AE1-070 2	13.29
938851	AE1-113 C O1	10.63
938852	AE1-113 E O1	37.7

938861	AE1-114 C O1	2.73
938862	AE1-114 E O1	9.32
939051	AE1-134 1	0.99
939061	AE1-134 2	0.99
939321	AE1-163 C O1	7.51
939322	AE1-163 E O1	46.12
939683	AE1-198 BAT	25.4
LTF	BLUEG	2.75
294401	BSHIL;1U E	6.61
294410	BSHIL;2U E	6.61
LTF	CARR	0.35
LTF	CATAWBA	0.05
LTF	CBM-S1	2.18
LTF	CBM-W1	18.75
LTF	CBM-W2	46.22
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
274859	EASYR;U1 E	7.77
274860	EASYR;U2 E	7.77
274735	ELWOOD EC;4P	5.44
274728	ELWOOD EC;5P	5.47
274730	ELWOOD EC;6P	5.47
274732	ELWOOD EC;7P	5.47
274734	ELWOOD EC;8P	5.47
274736	ELWOOD EC;9P	5.47
LTF	G-007	0.96
LTF	GIBSON	< 0.01
LTF	HAMLET	0.13
990901	L-005 E	9.74
LTF	MEC	29.62
293516	O-009 E1	6.66
293517	O-009 E2	3.38
293518	O-009 E3	3.72
293715	O-029 E	7.12
293716	O-029 E	3.9
293717	O-029 E	3.59
293771	O-035 E	4.74
LTF	O-066	6.14
293644	O22 E1	24.88
293645	O22 E2	48.31
290021	O50 E	25.78
294763	P-046 E	6.41
270859	PWR VTR EC;R	8.02
LTF	RENSSELAER	0.27
274722	S-055 E	7.48

274861	TOP CROP ;1U	1.1
274862	TOP CROP ;2U	2.13
LTF	TRIMBLE	0.33
LTF	WEC	5.43
915011	Y3-013 1	2.49
915021	Y3-013 2	2.49
915031	Y3-013 3	2.49
916211	Z1-072 E	3.59
916522	Z1-108 E	1.61
919221	AA1-146	12.77
919581	AA2-030	12.77
920272	AA2-123 E	1.61
930481	AB1-089	44.23
930741	AB1-122 1O1	65.11
930751	AB1-122 2O1	95.98
924471	AB2-096	28.08
925581	AC1-033 C	1.08
925582	AC1-033 E	7.24
926331	AC1-110 1	1.21
926341	AC1-110 2	1.21
927511	AC1-113 1	0.8
927521	AC1-113 2	0.8
926431	AC1-114	1.6
927451	AC1-142A 1	3.7
927461	AC1-142A 2	3.74
927091	AC1-204 1	115.58
927101	AC1-204 2	114.93
927201	AC1-214 C O1	1.52
927202	AC1-214 E O1	4.83

Appendix 6

(CE - AEP) The GREENACRE; T-05OLIVE 345 kV line (from bus 270771 to bus 243229 ckt 1) loads from 112.54% to 112.96% (AC power flow) of its emergency rating (971 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#2978_05DUMONT 765_B'. This project contributes approximately 59.94 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#2978_05DUMONT 765_B'

OPEN BRANCH FROM BUS 243206 TO BUS 243207 CKT 1 / 243206
05DUMONT 765 243207 05GRNTWN 765 1

OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206
05DUMONT 765 270644 WILTON ; 765 1

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	1.59
932891	AC2-115 2	1.59
932921	AC2-116	0.56
933411	AC2-154 C	1.69
933412	AC2-154 E	2.76
933911	AD1-013 C	1.23
933912	AD1-013 E	1.96
933931	AD1-016 C	0.62
933932	AD1-016 E	1.02
934051	AD1-031 C O1	1.88
934052	AD1-031 E O1	3.07
934101	AD1-039 1	4.69
934111	AD1-039 2	4.9
934431	AD1-067 C	0.09
934432	AD1-067 E	0.37
934701	AD1-098 C O1	4.6
934702	AD1-098 E O1	3.35
934721	AD1-100 C	13.01
934722	AD1-100 E	60.71
934871	AD1-116 C	0.62
934872	AD1-116 E	1.01
934971	AD1-129 C	0.61
934972	AD1-129 E	0.4
935001	AD1-133 C O1	13.81
935002	AD1-133 E O1	9.21
936291	AD2-038 C O1	1.55
936292	AD2-038 E O1	10.38
936371	AD2-047 C O1	3.03

936372	AD2-047 E O1	14.78
936461	AD2-060	1.78
936511	AD2-066 C O1	5.6
936512	AD2-066 E O1	3.73
936791	AD2-102 C	9.47
936792	AD2-102 E	6.31
937001	AD2-134 C	1.73
937002	AD2-134 E	6.9
937311	AD2-172 C	1.65
937312	AD2-172 E	2.27
937401	AD2-194 1	5.27
937411	AD2-194 2	5.27
937531	AD2-214 C	2.94
937532	AD2-214 E	1.96
938511	AE1-070 1	6.19
938521	AE1-070 2	5.66
938851	AE1-113 C O1	5.39
938852	AE1-113 E O1	19.1
938861	AE1-114 C O1	2.63
938862	AE1-114 E O1	8.96
939051	AE1-134 1	0.91
939061	AE1-134 2	0.91
939321	AE1-163 C O1	3.9
939322	AE1-163 E O1	23.94
939351	AE1-166 C O1	6.87
939352	AE1-166 E O1	6.34
939401	AE1-172 C O1	3.5
939402	AE1-172 E O1	16.4
939631	AE1-193 C O1	10.91
939632	AE1-193 E O1	73.04
939641	AE1-194 C	10.91
939642	AE1-194 E	73.04
939651	AE1-195 C	10.91
939652	AE1-195 E	73.04
939681	AE1-198 C O1	32.41
939682	AE1-198 E O1	27.54
939741	AE1-205 C O1	5.79
939742	AE1-205 E O1	8.
940101	AE1-252 C O1	7.02
940102	AE1-252 E O1	4.68
LTF	BLUEG	2.63
294401	BSHIL;1U E	5.66
294410	BSHIL;2U E	5.66
LTF	CARR	0.48
LTF	CATAWBA	0.15

274890	CAYUG;1U E	8.97
274891	CAYUG;2U E	8.97
LTF	CBM-S1	1.49
LTF	CBM-W1	20.35
LTF	CBM-W2	39.71
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
LTF	CIN	0.23
274751	CRETE EC ;1U	2.22
274752	CRETE EC ;2U	2.22
274753	CRETE EC ;3U	2.22
274754	CRETE EC ;4U	2.22
274859	EASYR;U1 E	7.34
274860	EASYR;U2 E	7.34
LTF	G-007	1.33
290051	GSG-6; E	6.98
LTF	HAMLET	0.3
953871	J847	2.7
275149	KEMPTON ;1E	12.46
990901	L-005 E	8.29
290108	LEEDK;1U E	16.23
LTF	MEC	25.26
293061	N-015 E	10.29
293516	O-009 E1	6.08
293517	O-009 E2	3.09
293518	O-009 E3	3.4
293715	O-029 E	6.5
293716	O-029 E	3.56
293717	O-029 E	3.28
293771	O-035 E	4.23
LTF	O-066	8.55
293644	O22 E1	7.28
293645	O22 E2	14.13
290021	O50 E	13.06
294392	P-010 E	13.07
294763	P-046 E	6.27
274888	PILOT HIL;1E	12.46
270859	PWR VTR EC;R	8.1
LTF	RENSSELAER	0.38
274722	S-055 E	7.56
295111	SUBLETTE E	1.82
LTF	TRIMBLE	0.32
LTF	WEC	5.35
295109	WESTBROOK E	3.74
910542	X3-005 E	0.52

915011	Y3-013 1	2.52
915021	Y3-013 2	2.52
915031	Y3-013 3	2.52
916211	Z1-072 E	3.2
916221	Z1-073 E	3.6
916502	Z1-106 E1	0.85
916504	Z1-106 E2	0.85
916512	Z1-107 E	1.72
916522	Z1-108 E	1.67
917502	Z2-087 E	11.99
918052	AA1-018 E	10.68
919221	AA1-146	11.73
919581	AA2-030	11.73
920272	AA2-123 E	1.64
930481	AB1-089	44.01
930501	AB1-091 O1	49.4
930741	AB1-122 1O1	47.85
930751	AB1-122 2O1	50.02
924041	AB2-047 C O1	2.24
924042	AB2-047 E O1	14.99
924471	AB2-096	28.36
925302	AB2-191 E	0.92
925581	AC1-033 C	0.93
925582	AC1-033 E	6.21
926311	AC1-109 1	1.28
926321	AC1-109 2	1.28
926331	AC1-110 1	1.27
926341	AC1-110 2	1.27
926351	AC1-111 1	0.51
926361	AC1-111 2	0.51
926371	AC1-111 3	0.51
926381	AC1-111 4	0.51
926391	AC1-111 5	0.51
926401	AC1-111 6	0.51
927511	AC1-113 1	0.8
927521	AC1-113 2	0.8
926431	AC1-114	1.59
927451	AC1-142A 1	2.85
927461	AC1-142A 2	2.85
926821	AC1-168 C O1	0.76
926822	AC1-168 E O1	5.11
927091	AC1-204 1	49.02
927101	AC1-204 2	48.98
927201	AC1-214 C O1	1.36
927202	AC1-214 E O1	4.32

Appendix 7

(CE - MISO NIPS) The ST JOHN ; T-17GREEN_ACRE 345 kV line (from bus 270886 to bus 255104 ckt 1) loads from 122.96% to 132.31% (AC power flow) of its emergency rating (1091 MVA) for the line fault with failed breaker contingency outage of 'COMED_P4_023-65-BT2-3__'. This project contributes approximately 104.3 MW to the thermal violation.

CONTINGENCY 'COMED_P4_023-65-BT2-3__'

TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765
05DUMONT 765

TRIP BRANCH FROM BUS 270607 TO BUS 270630 CKT 1 / COLLI; 765
PLANO; 765

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	1.78
932891	AC2-115 2	1.78
932921	AC2-116	0.62
933411	AC2-154 C	1.75
933412	AC2-154 E	2.85
933911	AD1-013 C	1.38
933912	AD1-013 E	2.2
933931	AD1-016 C	0.7
933932	AD1-016 E	1.14
934101	AD1-039 1	5.22
934111	AD1-039 2	5.55
934431	AD1-067 C	0.1
934432	AD1-067 E	0.41
934701	AD1-098 C O1	5.14
934702	AD1-098 E O1	3.76
934721	AD1-100 C	14.35
934722	AD1-100 E	66.95
934871	AD1-116 C	0.68
934872	AD1-116 E	1.11
934971	AD1-129 C	0.68
934972	AD1-129 E	0.45
935001	AD1-133 C O1	15.25
935002	AD1-133 E O1	10.17
936291	AD2-038 C O1	1.72
936292	AD2-038 E O1	11.52
936371	AD2-047 C O1	3.13
936372	AD2-047 E O1	15.27
936461	AD2-060	1.84

936511	AD2-066 C O1	6.23
936512	AD2-066 E O1	4.15
936791	AD2-102 C	10.6
936792	AD2-102 E	7.07
937001	AD2-134 C	1.94
937002	AD2-134 E	7.73
937311	AD2-172 C	1.84
937312	AD2-172 E	2.54
937401	AD2-194 1	5.97
937411	AD2-194 2	5.96
938511	AE1-070 1	7.01
938521	AE1-070 2	6.4
938851	AE1-113 C O1	6.08
938852	AE1-113 E O1	21.55
938861	AE1-114 C O1	2.94
938862	AE1-114 E O1	10.02
939051	AE1-134 1	1.02
939061	AE1-134 2	1.02
939321	AE1-163 C O1	4.32
939322	AE1-163 E O1	26.57
939351	AE1-166 C O1	7.64
939352	AE1-166 E O1	7.05
939401	AE1-172 C O1	3.83
939402	AE1-172 E O1	17.91
939631	AE1-193 C O1	18.99
939632	AE1-193 E O1	127.09
939641	AE1-194 C	18.99
939642	AE1-194 E	127.09
939651	AE1-195 C	18.99
939652	AE1-195 E	127.09
939681	AE1-198 C O1	56.38
939682	AE1-198 E O1	47.91
940101	AE1-252 C O1	7.67
940102	AE1-252 E O1	5.12
LTF	BLUEG	4.79
274654	BRAIDWOOD;1U	22.
274655	BRAIDWOOD;2U	21.06
LTF	CALDERWOOD	0.06
LTF	CANNELTON	0.09
LTF	CARR	0.54
LTF	CATAWBA	0.21
274890	CAYUG;1U E	8.79
274891	CAYUG;2U E	8.79
LTF	CBM-S1	0.91
LTF	CBM-W1	20.43

LTF	CBM-W2	38.51
LTF	CHEOAH	0.06
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
274751	CRETE EC ;1U	3.87
274752	CRETE EC ;2U	3.87
274753	CRETE EC ;3U	3.87
274754	CRETE EC ;4U	3.87
274859	EASYR;U1 E	8.22
274860	EASYR;U2 E	8.22
LTF	G-007	1.52
LTF	GIBSON	0.05
290051	GSG-6; E	7.82
LTF	HAMLET	0.4
275149	KEMPTON ;1E	12.88
274704	KENDALL ;1C	3.25
274705	KENDALL ;1S	2.17
274706	KENDALL ;2C	3.25
274707	KENDALL ;2S	2.17
274661	LASCO STA;2U	20.32
290108	LEEDK;1U E	18.19
LTF	MEC	27.81
293061	N-015 E	11.68
293516	O-009 E1	6.79
293517	O-009 E2	3.45
293518	O-009 E3	3.8
293715	O-029 E	7.26
293716	O-029 E	3.98
293717	O-029 E	3.66
LTF	O-066	9.77
293644	O22 E1	8.54
293645	O22 E2	16.59
290021	O50 E	14.73
294392	P-010 E	14.83
294763	P-046 E	7.02
274888	PILOT HIL;1E	12.88
270859	PWR VTR EC;R	9.09
LTF	RENSSELAER	0.43
274722	S-055 E	8.49
295111	SUBLETTE E	2.04
274861	TOP CROP ;1U	0.38
274862	TOP CROP ;2U	0.73
LTF	TRIMBLE	0.56
LTF	WEC	6.
295109	WESTBROOK E	4.19

274687	WILL CNTY;4U	9.27
915011	Y3-013 1	2.83
915021	Y3-013 2	2.83
915031	Y3-013 3	2.83
916221	Z1-073 E	4.03
916502	Z1-106 E1	0.95
916504	Z1-106 E2	0.95
916512	Z1-107 E	1.86
916522	Z1-108 E	1.87
918052	AA1-018 E	11.7
919221	AA1-146	13.12
919581	AA2-030	13.12
920272	AA2-123 E	1.84
930481	AB1-089	49.32
930501	AB1-091 O1	50.89
930741	AB1-122 1O1	53.27
930751	AB1-122 2O1	56.61
924471	AB2-096	31.8
925302	AB2-191 E	1.04
926311	AC1-109 1	1.43
926321	AC1-109 2	1.43
926331	AC1-110 1	1.43
926341	AC1-110 2	1.43
926351	AC1-111 1	0.57
926361	AC1-111 2	0.57
926371	AC1-111 3	0.57
926381	AC1-111 4	0.57
926391	AC1-111 5	0.57
926401	AC1-111 6	0.57
927511	AC1-113 1	0.89
927521	AC1-113 2	0.89
926431	AC1-114	1.78
927451	AC1-142A 1	3.22
927461	AC1-142A 2	3.22
926821	AC1-168 C O1	0.85
926822	AC1-168 E O1	5.72
927091	AC1-204 1	55.48
927101	AC1-204 2	55.4

Appendix 8

(CE - MISO NIPS) The CRETE EC ;BP-17STJOHN 345 kV line (from bus 274750 to bus 255112 ckt 1) loads from 149.16% to 162.01% (AC power flow) of its emergency rating (1399 MVA) for the line fault with failed breaker contingency outage of 'COMED_P4_112-65-BT4-5__'. This project contributes approximately 178.46 MW to the thermal violation.

CONTINGENCY 'COMED_P4_112-65-BT4-5__'

TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765
05DUMONT 765

TRIP BRANCH FROM BUS 275233 TO BUS 270644 CKT 1 / WILTO;4M 345
WILTO; 765

TRIP BRANCH FROM BUS 275233 TO BUS 270927 CKT 1 / WILTO;4M 345
WILTO; R 345

TRIP BRANCH FROM BUS 275233 TO BUS 275333 CKT 1 / WILTO;4M 345
WILTO;4C 33

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	2.5
932891	AC2-115 2	2.5
932921	AC2-116	0.87
933411	AC2-154 C	2.32
933412	AC2-154 E	3.78
933911	AD1-013 C	1.92
933912	AD1-013 E	3.07
933931	AD1-016 C	0.98
933932	AD1-016 E	1.59
934101	AD1-039 1	7.31
934111	AD1-039 2	7.82
934431	AD1-067 C	0.14
934432	AD1-067 E	0.58
934701	AD1-098 C O1	7.19
934702	AD1-098 E O1	5.25
934721	AD1-100 C	19.9
934722	AD1-100 E	92.88
934871	AD1-116 C	0.94
934872	AD1-116 E	1.54
934971	AD1-129 C	0.95
934972	AD1-129 E	0.63
935001	AD1-133 C O1	21.27
935002	AD1-133 E O1	14.18
936291	AD2-038 C O1	2.4

936292	AD2-038 E O1	16.06
936371	AD2-047 C O1	4.15
936372	AD2-047 E O1	20.26
936461	AD2-060	2.44
936511	AD2-066 C O1	8.7
936512	AD2-066 E O1	5.8
936791	AD2-102 C	14.83
936792	AD2-102 E	9.89
937001	AD2-134 C	2.71
937002	AD2-134 E	10.8
937311	AD2-172 C	2.58
937312	AD2-172 E	3.56
937401	AD2-194 1	8.41
937411	AD2-194 2	8.4
938511	AE1-070 1	9.89
938521	AE1-070 2	9.03
938851	AE1-113 C O1	8.56
938852	AE1-113 E O1	30.35
938861	AE1-114 C O1	4.11
938862	AE1-114 E O1	14.02
939051	AE1-134 1	1.43
939061	AE1-134 2	1.43
939321	AE1-163 C O1	6.03
939322	AE1-163 E O1	37.03
939351	AE1-166 C O1	10.65
939352	AE1-166 E O1	9.83
939401	AE1-172 C O1	5.29
939402	AE1-172 E O1	24.78
939631	AE1-193 C O1	32.49
939632	AE1-193 E O1	217.46
939641	AE1-194 C	32.49
939642	AE1-194 E	217.46
939651	AE1-195 C	32.49
939652	AE1-195 E	217.46
939681	AE1-198 C O1	96.48
939682	AE1-198 E O1	81.98
940101	AE1-252 C O1	10.61
940102	AE1-252 E O1	7.08
LTF	BLUEG	7.1
274654	BRAIDWOOD;1U	31.29
274655	BRAIDWOOD;2U	29.94
LTF	CALDERWOOD	0.11
LTF	CANNELTON	0.16
LTF	CARR	0.77
LTF	CATAWBA	0.32

LTF	CBM-S1	1.08
LTF	CBM-W1	26.5
LTF	CBM-W2	52.14
LTF	CHEOAH	0.11
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
274751	CRETE EC ;1U	6.62
274752	CRETE EC ;2U	6.62
274753	CRETE EC ;3U	6.62
274754	CRETE EC ;4U	6.62
274859	EASYR;U1 E	11.49
274860	EASYR;U2 E	11.49
LTF	G-007	2.16
LTF	GIBSON	0.09
290051	GSG-6; E	10.92
LTF	HAMLET	0.59
275149	KEMPTON ;1E	17.08
274704	KENDALL ;1C	4.59
274705	KENDALL ;1S	3.06
274706	KENDALL ;2C	4.59
274707	KENDALL ;2S	3.06
274661	LASCO STA;2U	28.64
290108	LEEDK;1U E	25.39
LTF	MEC	38.8
293061	N-015 E	16.42
293516	O-009 E1	9.5
293517	O-009 E2	4.82
293518	O-009 E3	5.31
293715	O-029 E	10.16
293716	O-029 E	5.57
293717	O-029 E	5.12
LTF	O-066	13.85
293644	O22 E1	12.32
293645	O22 E2	23.92
290021	O50 E	20.75
294392	P-010 E	20.85
294763	P-046 E	9.81
274888	PILOT HIL;1E	17.08
270859	PWR VTR EC;R	12.71
LTF	RENSSELAER	0.61
274722	S-055 E	11.9
295111	SUBLETTE E	2.84
274861	TOP CROP ;1U	0.54
274862	TOP CROP ;2U	1.05
LTF	TRIMBLE	0.83

<i>LTF</i>	<i>WEC</i>	<i>8.41</i>
295109	WESTBROOK E	5.85
274687	WILL CNTY;4U	13.05
915011	Y3-013 1	3.97
915021	Y3-013 2	3.97
915031	Y3-013 3	3.97
916221	Z1-073 E	5.64
916502	Z1-106 E1	1.33
916504	Z1-106 E2	1.33
916512	Z1-107 E	2.55
916522	Z1-108 E	2.62
918052	AA1-018 E	16.17
919221	AA1-146	18.35
919581	AA2-030	18.35
920272	AA2-123 E	2.57
930481	AB1-089	68.97
930501	AB1-091 O1	67.31
930741	AB1-122 1O1	74.6
930751	AB1-122 2O1	79.79
924471	AB2-096	44.48
925302	AB2-191 E	1.45
926311	AC1-109 1	2.
926321	AC1-109 2	2.
926331	AC1-110 1	2.
926341	AC1-110 2	2.
926351	AC1-111 1	0.8
926361	AC1-111 2	0.8
926371	AC1-111 3	0.8
926381	AC1-111 4	0.8
926391	AC1-111 5	0.8
926401	AC1-111 6	0.8
927511	AC1-113 1	1.25
927521	AC1-113 2	1.25
926431	AC1-114	2.5
927451	AC1-142A 1	4.53
927461	AC1-142A 2	4.53
926821	AC1-168 C O1	1.19
926822	AC1-168 E O1	8.01
927091	AC1-204 1	78.25
927101	AC1-204 2	78.11

Appendix 9

(CE - CE) The E FRANKFO; B-CRETE EC ;BP 345 kV line (from bus 270728 to bus 274750 ckt 1) loads from 104.74% to 107.95% (**DC power flow**) of its emergency rating (1399 MVA) for the single line contingency outage of 'AEP_P1-2_#695A'. This project contributes approximately 44.97 MW to the thermal violation.

CONTINGENCY 'AEP_P1-2_#695A'

OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206

05DUMONT 765 270644 WILTON ; 765 1

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
934431	AD1-067 C	0.1
934432	AD1-067 E	0.44
934721	AD1-100 C	15.33
934722	AD1-100 E	71.56
935141	AD1-148	2.68
936291	AD2-038 C O1	1.84
936292	AD2-038 E O1	12.29
936371	AD2-047 C O1	3.17
936372	AD2-047 E O1	15.47
936972	AD2-131 E	2.79
937001	AD2-134 C	2.07
937002	AD2-134 E	8.26
937211	AD2-159 C	2.
937212	AD2-159 E	9.38
938851	AE1-113 C O1	7.2
938852	AE1-113 E O1	22.64
938861	AE1-114 C O1	2.87
938862	AE1-114 E O1	10.98
939321	AE1-163 C O1	4.61
939322	AE1-163 E O1	28.34
939401	AE1-172 C	4.05
939402	AE1-172 E	18.99
939682	AE1-198 E	44.97
274857	BIG SKY ;U1	1.1
274858	BIG SKY ;U2	1.1
274877	BISHOP HL;1U	0.84
274878	BISHOP HL;2U	0.84
294401	BSHIL;1U E	3.36
294410	BSHIL;2U E	3.36
274848	CAMP GROVE;RU	1.23
274890	CAYUG;1U E	5.21

274891	CAYUG;2U E	5.21
274863	CAYUGA RI;1U	1.3
274864	CAYUGA RI;2U	1.3
274849	CRESCENT ;1U	0.41
274859	EASYR;U1 E	4.41
274860	EASYR;U2 E	4.41
274856	ECOGROVE ;U1	0.94
950751	G858	1.93
274871	GR RIDGE ;2U	2.01
274847	GR RIDGE ;BU	1.58
274855	GSG-6 ;RU	1.05
290051	GSG-6; E	4.2
950591	H008	1.96
950761	H071	2.03
950671	J112	0.28
950141	J395	6.82
950181	J407	10.02
950221	J416	10.09
950491	J443	2.24
950501	J449	10.8
952021	J614	3.57
953201	J715	6.91
954701	J844	8.89
953921	J855	5.31
954091	J873	10.41
953011	J885	3.21
954301	J898	5.41
954511	J926	5.82
954741	J928	5.92
275149	KEMPTON ;1E	6.55
990901	L-005 E	4.9
274872	LEE DEKAL;1U	2.34
290108	LEEDK;1U E	9.77
274850	MENDOTA H;RU	0.29
274879	MINONK ;1U	2.
293061	N-015 E	6.34
293513	O-009 C1	0.9
293514	O-009 C2	0.46
293515	O-009 C3	0.51
293516	O-009 E1	3.61
293517	O-009 E2	1.83
293518	O-009 E3	2.02
293712	O-029 C	0.96
293713	O-029 C	0.53
293714	O-029 C	0.49

293715	O-029 E	3.86
293716	O-029 E	2.12
293717	O-029 E	1.95
293771	O-035 E	2.51
293644	O22 E1	4.78
293645	O22 E2	9.28
290021	O50 E	8.
294392	P-010 E	8.05
294763	P-046 E	3.77
274888	PILOT HIL;1E	6.56
274887	PILOT HIL;1U	1.64
274881	PLEAS RDG;2U	1.64
274851	PROVIDENC;RU	0.63
290261	S-027 E	5.48
290265	S-028 E	5.48
295110	SUBLETTE C	0.15
274861	TOP CROP ;1U	1.2
274862	TOP CROP ;2U	2.32
274853	TWINGROVE;U1	1.37
274854	TWINGROVE;U2	1.37
905081	W4-005 C	1.39
905082	W4-005 E	6.58
295108	WESTBROOK C	0.34
909052	X2-022 E	10.24
916211	Z1-072 E	1.9
916221	Z1-073 E	2.17
917501	Z2-087 C	2.06
917502	Z2-087 E	13.81
918051	AA1-018 C	1.85
918052	AA1-018 E	12.38
920272	AA2-123 E	2.46
924041	AB2-047 C O1	2.55
924042	AB2-047 E O1	17.04
924261	AB2-070 C O1	1.46
924262	AB2-070 E O1	9.78
925301	AB2-191 C	0.6
925302	AB2-191 E	0.83
925581	AC1-033 C	1.1
925582	AC1-033 E	7.34
925771	AC1-053 C	1.45
925772	AC1-053 E	9.69
926821	AC1-168 C O1	0.91
926822	AC1-168 E O1	6.1
926841	AC1-171 C O1	0.75
926842	AC1-171 E O1	4.98

<i>927201</i>	<i>AC1-214 C O1</i>	<i>1.61</i>
<i>927202</i>	<i>AC1-214 E O1</i>	<i>5.11</i>

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